AthDGC at CIDL 2026

Civis Diachronic Linguistics · Athens · CIVIS BIP

Nikolaos Lavidas

June 15, 2026

AthDGC, Athens-PROIEL (the dependency-treebank (a collection of sentences whose grammatical structure has been analysed and stored) standard for early Indo-European languages, developed at Oslo)

Diachronic Greek treebank · Indo-European cross-lingual alignment (matching corresponding words or sentences between texts in different languages) Athens Digital Glossa Chronos Research Network

HFRI Project No. 20577 · Greece 2.0 NRRP · CVL-CDSAML

Why this matters at CIDL

Diachronic linguistics needs:

- continuous coverage across periods
- aligned witnesses across languages
- uniform syntactic annotation
- machine-queryable argument structure (the set of obligatory and optional partners a verb requires, e.g. subject, object, oblique)

No single existing resource provides all four. AthDGC does.

Coverage (v0.4)

Period	Years	Annotated Greek rows
Archaic	8th–6th c. BC	Homer, Hesiod, Pindar, Sappho
Classical	5th–4th c. BC	Tragedy + comedy + history + philosophy + oratory
Koine	3rd c. BC – 4th c. AD	New Testament, Plutarch, Strabo, Lucian
Late Antique	4th–7th c. AD	Eusebius, Basil, Chrysostom, Procopius
Byzantine	7th–12th c. AD	Psellos, Anna Komnene, John of Damascus, Niketas
Late Byzantine	13th–15th c. AD	*in ingestion*
Early Modern	16th–18th c. AD	*in ingestion*

Cross-lingual alignment

NT verse-level cross-alignment to:

- **Latin** (Vulgate), Clementine Vulgate NT aligned
- **Gothic** (Wulfila), *in ingestion*
- **Old Church Slavonic** (Marianus), *in ingestion*
- **Classical Armenian**, PROIEL-style workflow in development

Method: LaBSE (a multilingual sentence-embedding model that maps sentences from many languages into a common vector space) sentence embedding (a numeric vector that represents the meaning of a sentence) + AwesomeAlign (a word-alignment procedure that uses multilingual-BERT attention to match words across translations) word-level (mBERT attention).

Argument structure

For every VERB / AUX token AthDGC extracts:

- subject (`sub`, including raised `xobj` / `nonsub` patterns)
- direct object (`obj`)
- indirect / oblique args (`iobj`, `obl`)
- vocative addressee (`voc`)
- voice (active / middle / passive)
- aspect (perfective / imperfective)
- tense, mood, verb-form

Stored per row in a column accessible to graph queries over the cross-lingual alignment edges.

Example query

> *"Show every Greek aorist transitive verb with an accusative object whose Latin Vulgate counterpart is a passive periphrastic"*

Translates to a Neo4j (a graph database that stores data as nodes and connections rather than as tables) Cypher query traversing the `(Token)-[TRANSLATED_AS]->(Token)` edges where source carries `voice=Act,aspect=Perf,obj=Acc` and target carries `voice=Pass,verb_form=Periphrastic`.

Open infrastructure

- **Source code:** https://github.com/AthDGC/Diachronic-Linguistics-Platform (Apache-2.0 (a permissive open-source licence widely used for software))
- **Dataset DOI (Digital Object Identifier, a permanent web link for a published artefact):** 10.5281/zenodo (an open research-data repository hosted at CERN that mints permanent DOIs).20439182 (CC-BY-4.0 (the Creative Commons Attribution 4.0 licence, which permits reuse with credit))
- **Showcase:** athdgc.github.io
- **Hugging Face (a public hosting platform for machine-learning models and datasets) checkpoints (saved snapshots of a trained model):** mirror at https://huggingface.co/AthDGC in setup; released at v0.5
- **Launch report:** GlossaContactLab Working Papers, NKUA digital edition

Team

PI: Prof. Nikolaos Lavidas (NKUA) **International:** Prof. Dag Haug (Oslo, PROIEL Project Director); Prof. Leonid Kulikov (Ghent; Diachronic typology, valency (the number and type of arguments a verb takes) questionnaires) **NKUA:** Prof. Emerita Kiki Nikiforidou; Dr. Vassiliki Geka, Dr. Vassileios Symeonidis, Dr. Theodoros Michalareas (Post-Doctoral Researchers); Sofia Chionidi, Anastasia Tsiropina, Eleni Plakoutsi (PhD Candidates); Evangelos Argyropoulos (Research Assistant)

Funded by HFRI (Project No. 20577) · Greece 2.0 NRRP Compute: GRNET ARIS (the Greek national high-performance computing cluster, run by GRNET) (pa260305)

Discussion

What CIDL participants can do with AthDGC today:

1. Query the corpus via the public Neo4j endpoint
2. Pull period-specific JSONL (a plain-text data format that stores one record per line) partitions from Hugging Face
3. Browse the PROIEL XML 2.0 (the file format that stores each sentence as a tree of word-by-word grammatical relations, developed at Oslo) samples on `athdgc.github.io`
4. Flag annotation errors via the showcase review toolbar
5. Cite version-pinned datasets via the version DOIs

Thank you

athdgc.github.io · github.com/AthDGC 10.5281/zenodo.20439182

Funding

Funded by the **Hellenic Foundation for Research and Innovation (HFRI)** under the 3rd Call for HFRI Research Projects to support Post-Doctoral Researchers, **Project No. 20577**; with complementary support from the **Greece 2.0 National Recovery and Resilience Plan**. Compute supplied by **GRNET ARIS** (Greek national HPC (high-performance computing, i.e. a cluster of fast machines used for heavy computation)), allocation pa260305.